main — Discovery of Y-chromosome deletion guides

Repetitive Y-specific targets for chromosome shredding and centromere targeting

a Enumerate repeated, PAM-compatible chrY sequences

HG002 T2T-Y

62,460,029 bp

CP086569.2

Count identical 23-mers

23-nt protospacer–PAM sequence

5′ | 20-nt protospacer | NGG | 3′

5′ | CCN | 20-nt protospacer | 3′

Recognise either PAM orientation

b Filter recurrent Y target sequences

Recurrent, PAM-compatible

13,503

distinct 23-mers

≥18 stored Y occurrences

No exact matches outside chrY

8,104

distinct 23-mers

chr1–22, chrX and chrM

High repeat abundance

7,470

distinct 23-mers

>20 occurrences on chrY

c Assess candidate guides (20-nt spacers)

Targeting pattern

Exact-match abundance and regional distribution across both orientations

Near-match profile

1–4 mismatches and CFD
Summarise chrY near-matches separately

Predicted activity

DeepSpCas9 predictions where available

d Initial selection: five complementary candidates

Yq repeats	Centromere		Yp	
Q53	CN	CP	P47	P51
532	53	54	47	51

Exact chrY matches; all five have zero exact non-Y matches

Counts sum forward- and reverse-orientation matches